

Performance Regression Detection

Complete Test Execution Status Report - TC-01 to TC-20 + EC-01 to EC-20

Metric	Final Result
Total validations	40
Main test cases	20 / 20 PASS
Edge cases	20 / 20 PASS
Failed after final retest	0
Overall result	PASS

Scope: Final validation of the Performance Regression Detection feature, covering the 20 functional test cases and the 20 edge cases supplied during this testing session.

Evidence policy: TC evidence pages use the captured Swagger screenshots from the session. EC evidence uses the EC-01 Swagger screenshot where available and otherwise preserves the exact request/result values recorded in the conversation/terminal output; missing screenshots are not fabricated.

Resolved during testing: TC-14 and EC-01, EC-02, EC-10, EC-13 and EC-20 required targeted code changes and successful retests before being marked PASS.

Final summary: 20/20 main test cases PASS, 20/20 edge cases PASS, 40/40 total PASS.

Main Test Case Results

TC ID	Scenario	Status	Final Result
TC-01	No performance change	PASS	Equal old/current execution times did not trigger regression.
TC-02	Small performance increase	PASS	Small increase below configured threshold was correctly ignored.
TC-03	Significant slowdown	PASS	5x slowdown crossed the threshold.
TC-04	Exactly at regression threshold	PASS	Exact 2x boundary consistently treated as regression.
TC-05	Slightly above threshold	PASS	Barely-above-threshold degradation was correctly detected.
TC-06	Very large slowdown	PASS	Very large slowdown was detected.
TC-07	Query becomes faster	PASS	Improvement did not create a false regression.
TC-08	Multiple executions	PASS	Historical/current aggregates were calculated correctly.
TC-09	Same query repeated	PASS	Repeated executions were compared by the defined baseline method.
TC-10	Multiple queries	PASS	Only affected query Q2 was flagged.
TC-11	Stable query	PASS	Natural near-stability remained below threshold.
TC-12	Gradual slowdown	PASS	Configured rule detected the progressive slowdown.
TC-13	Sudden slowdown	PASS	Sudden 8x degradation was detected.
TC-14	Performance improvement after optimization	PASS	Initial response lacked explicit improvement flag; after fix and retest, improvement was recorded.
TC-15	Regression report	PASS	Report exposed query, old/new times, slowdown and status.
TC-16	Historical baseline available	PASS	Current result correctly compared against historical baseline values supplied to the API.
TC-17	Different query plans	PASS	Plan change was recorded and performance degradation crossed the rule.
TC-18	Data growth causes slowdown	PASS	Measured slowdown was correctly detected for the data-growth scenario.
TC-19	Index removed	PASS	Index-removal scenario was correctly flagged.
TC-20	Statistics / plan change	PASS	Statistics/plan-change scenario crossed the degradation rule.

Edge Case Results

EC ID	Scenario	Status	Final Result
EC-01	No historical data	PASS	First-time run now establishes baseline without claiming regression.
EC-02	Historical baseline = 0	PASS	Zero baseline no longer crashes or divides by zero.
EC-03	Negative execution time	PASS	Negative execution time rejected by input validation.
EC-04	Current time = baseline	PASS	No-change case correctly remained non-regression.
EC-05	Current time slightly higher	PASS	Tiny variation did not create a false positive.
EC-06	Exactly at threshold	PASS	Boundary consistently uses >= threshold.
EC-07	Just above threshold	PASS	Just-above-threshold case detected correctly.
EC-08	Query becomes faster	PASS	Improvement was explicitly recognized.
EC-09	Huge slowdown	PASS	Severe slowdown detected.
EC-10	Temporary spike	PASS	Single outlier no longer creates false regression.
EC-11	Noisy execution times	PASS	Natural noise did not trigger regression.
EC-12	Very small baseline	PASS	Small-baseline variation remained stable under the configured rule.
EC-13	Missing current time	PASS	Missing current measurement now returns an explicit no-decision result.
EC-14	Missing historical records	PASS	Current measurement established as baseline without regression claim.
EC-15	Very old historical data	PASS	Stale baseline was replaced according to configured freshness limit.
EC-16	Query text changed	PASS	Changed query text bypassed invalid historical comparison.
EC-17	Multiple executions at once	PASS	Concurrent requests returned separate correct results without cross-mixing.
EC-18	Database failure	PASS	Execution/database failure is separated from performance-regression classification.
EC-19	Configuration changes	PASS	Changed configuration was applied consistently.
EC-20	Data size changed dramatically	PASS	Regression detection and cause attribution are separated.

Overall Result: ALL 40 VALIDATIONS PASSED after final fixes/retests.

Detailed Test Cases - TC-01 to TC-20

TC-01 - No performance change STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=1, new=1, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; performance_ratio=1; regression_detected=false; result="No regression".
VALIDATION DETAIL	Equal old/current execution times did not trigger regression.
FINAL STATUS	PASS

TC-02 - Small performance increase STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=1, new=1.1, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; ratio=1.1; regression_detected=false; result="No regression".
VALIDATION DETAIL	Small increase below configured threshold was correctly ignored.
FINAL STATUS	PASS

TC-03 - Significant slowdown STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=1, new=5, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; ratio=5; regression_detected=true; result="Regression detected".
VALIDATION DETAIL	5x slowdown crossed the threshold.
FINAL STATUS	PASS

TC-04 - Exactly at regression threshold STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=2, new=4, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; ratio=2; boundary rule applied as ratio >= threshold.
VALIDATION DETAIL	Exact 2x boundary consistently treated as regression.
FINAL STATUS	PASS

TC-05 - Slightly above threshold STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=2, new=4.1, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; ratio=2.05; regression_detected=true.
VALIDATION DETAIL	Barely-above-threshold degradation was correctly detected.
FINAL STATUS	PASS

Detailed Test Cases - Continued

TC-06 - Very large slowdown STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=0.5, new=20, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; ratio=40; regression_detected=true.
VALIDATION DETAIL	Very large slowdown was detected.
FINAL STATUS	PASS

TC-07 - Query becomes faster STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=5, new=2, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; ratio=0.4; regression_detected=false; result="No regression".
VALIDATION DETAIL	Improvement did not create a false regression.
FINAL STATUS	PASS

TC-08 - Multiple executions STATUS: PASS

INPUT / TEST ACTION	POST /regression-history-check: old=[1,2,3], new=[2,3,4], threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; old_average=2; new_average=3; ratio=1.5; regression_detected=false.
VALIDATION DETAIL	Historical/current aggregates were calculated correctly.
FINAL STATUS	PASS

TC-09 - Same query repeated STATUS: PASS

INPUT / TEST ACTION	POST /regression-history-check: old=[1,1,1], new=[3,3,3], threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; old_average=1; new_average=3; ratio=3; regression_detected=true.
VALIDATION DETAIL	Repeated executions were compared by the defined baseline method.
FINAL STATUS	PASS

TC-10 - Multiple queries STATUS: PASS

INPUT / TEST ACTION	POST /multiple-regression-check: Q1 1->1, Q2 1->3, Q3 2->2, threshold=2.
OUTPUT / OBSERVED RESULT	Q1 ratio=1 false; Q2 ratio=3 true; Q3 ratio=1 false.
VALIDATION DETAIL	Only affected query Q2 was flagged.
FINAL STATUS	PASS

Detailed Test Cases - Continued

TC-11 - Stable query STATUS: PASS

INPUT / TEST ACTION	POST /regression-history-check: old=[2,2.1,2], new=[2.1,2,2.1], threshold=2.
OUTPUT / OBSERVED RESULT	old_average=2.0333; new_average=2.0667; ratio=1.01639; regression_detected=false.
VALIDATION DETAIL	Natural near-stability remained below threshold.
FINAL STATUS	PASS

TC-12 - Gradual slowdown STATUS: PASS

INPUT / TEST ACTION	POST /regression-history-check: old=[1,1.5,2,2.5], new=[5], threshold=2.
OUTPUT / OBSERVED RESULT	old_average=1.75; new_average=5; ratio=2.85714; regression_detected=true.
VALIDATION DETAIL	Configured rule detected the progressive slowdown.
FINAL STATUS	PASS

TC-13 - Sudden slowdown STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=1, new=8, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; ratio=8; regression_detected=true.
VALIDATION DETAIL	Sudden 8x degradation was detected.
FINAL STATUS	PASS

TC-14 - Performance improvement after optimization STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old=8, new=1, threshold=2.
OUTPUT / OBSERVED RESULT	Retest HTTP 200; ratio=0.125; regression_detected=false; improvement_detected=true; result="No regression".
VALIDATION DETAIL	Initial response lacked explicit improvement flag; after fix and retest, improvement was recorded.
FINAL STATUS	PASS

TC-15 - Regression report STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: query_name=Q1, old=1, new=5, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; query_name=Q1; slowdown_percent=400; ratio=5; regression_detected=true.
VALIDATION DETAIL	Report exposed query, old/new times, slowdown and status.
FINAL STATUS	PASS

Detailed Test Cases - Continued

TC-16 - Historical baseline available STATUS: PASS

INPUT / TEST ACTION	POST /regression-history-check: old=[1,1.1,0.9], new=[3], threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; old_average=1; new_average=3; ratio=3; regression_detected=true.
VALIDATION DETAIL	Current result correctly compared against historical baseline values supplied to the API.
FINAL STATUS	PASS

TC-17 - Different query plans STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old_plan="INDEX SCAN", new_plan="FULL TABLE SCAN", old=1, new=3, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; plan_changed=true; ratio=3; regression_detected=true.
VALIDATION DETAIL	Plan change was recorded and performance degradation crossed the rule.
FINAL STATUS	PASS

TC-18 - Data growth causes slowdown STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: query=Q_DATA_GROWTH, same plan, old=1, new=3, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; plan_changed=false; slowdown_percent=200; ratio=3; regression_detected=true.
VALIDATION DETAIL	Measured slowdown was correctly detected for the data-growth scenario.
FINAL STATUS	PASS

TC-19 - Index removed STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old_plan="INDEX SCAN", new_plan="FULL TABLE SCAN", old=1, new=3, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; plan_changed=true; slowdown_percent=200; ratio=3; regression_detected=true.
VALIDATION DETAIL	Index-removal scenario was correctly flagged.
FINAL STATUS	PASS

TC-20 - Statistics / plan change STATUS: PASS

INPUT / TEST ACTION	POST /regression-check: old_plan="INDEX SCAN", new_plan="FULL TABLE SCAN AFTER STATISTICS CHANGE", old=1, new=3, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; plan_changed=true; slowdown_percent=200; ratio=3; regression_detected=true.
VALIDATION DETAIL	Statistics/plan-change scenario crossed the degradation rule.
FINAL STATUS	PASS

Main Test Validation Notes

TC-14 fix/retest: The initial response showed no regression but did not record the improvement explicitly. An improvement_detected flag was added, and the retest passed with improvement_detected=true.

History/baseline cases: TC-08, TC-09, TC-11, TC-12 and TC-16 validated supplied historical/current execution arrays and comparison logic.

Plan/cause cases: TC-17 through TC-20 validated plan labels and measured degradation at the API level. These tests did not mutate the physical SQLite database plan/statistics/index during the session.

Final TC result: TC-01 through TC-20 = 20/20 PASS.

Detailed Edge Cases - EC-01 to EC-20

EC-01 - No historical data - Situation: First-time query runs STATUS: PASS

EXPECTED BEHAVIOUR	Do not claim regression; create baseline.
INPUT / TEST ACTION	POST /regression-history-check: old_execution_times=[], new_execution_times=[1], threshold=2.
OUTPUT / OBSERVED RESULT	Initial run: HTTP 500. After baseline-empty handling fix: HTTP 200; baseline_created=true; baseline=1; regression_detected=false; result="No regression - baseline created".
VALIDATION DETAIL	First-time run now establishes baseline without claiming regression.
FINAL STATUS	PASS

EC-02 - Historical baseline = 0 - Situation: Previous execution recorded as 0 STATUS: PASS

EXPECTED BEHAVIOUR	Avoid division-by-zero; use a safe comparison method.
INPUT / TEST ACTION	POST /regression-history-check: old=[0], new=[1], threshold=2.
OUTPUT / OBSERVED RESULT	After correcting return indentation/safe path: old_average=0.0; new_average=1.0; performance_ratio=1.0; regression_detected=false; result="No regression".
VALIDATION DETAIL	Zero baseline no longer crashes or divides by zero.
FINAL STATUS	PASS

EC-03 - Negative execution time - Situation: Invalid measurement STATUS: PASS

EXPECTED BEHAVIOUR	Reject/ignore invalid data.
INPUT / TEST ACTION	POST /regression-check: old_execution_time=-1, new_execution_time=1, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 422 validation response.
VALIDATION DETAIL	Negative execution time rejected by input validation.
FINAL STATUS	PASS

EC-04 - Current time = baseline - Situation: No change STATUS: PASS

EXPECTED BEHAVIOUR	No regression.
INPUT / TEST ACTION	POST /regression-check: old=2, new=2, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; regression_detected=false; result="No regression".
VALIDATION DETAIL	No-change case correctly remained non-regression.
FINAL STATUS	PASS

Detailed Edge Cases - Continued

EC-05 - Current time slightly higher - Situation: Tiny natural variation STATUS: PASS

EXPECTED BEHAVIOUR	Avoid false regression according to threshold.
INPUT / TEST ACTION	POST /regression-check: old=1, new=1.1, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; regression_detected=false; result="No regression".
VALIDATION DETAIL	Tiny variation did not create a false positive.
FINAL STATUS	PASS

EC-06 - Exactly at threshold - Situation: Current = baseline x threshold STATUS: PASS

EXPECTED BEHAVIOUR	Apply boundary rule consistently.
INPUT / TEST ACTION	POST /regression-check: old=2, new=4, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; regression_detected=true; result="Regression detected".
VALIDATION DETAIL	Boundary consistently uses >= threshold.
FINAL STATUS	PASS

EC-07 - Just above threshold - Situation: Current barely exceeds threshold STATUS: PASS

EXPECTED BEHAVIOUR	Correctly detect regression.
INPUT / TEST ACTION	POST /regression-check: old=2, new=4.01, threshold=2.
OUTPUT / OBSERVED RESULT	HTTP 200; regression_detected=true; result="Regression detected".
VALIDATION DETAIL	Just-above-threshold case detected correctly.
FINAL STATUS	PASS

EC-08 - Query becomes faster - Situation: Current < baseline STATUS: PASS

EXPECTED BEHAVIOUR	No regression; mark improvement if appropriate.
INPUT / TEST ACTION	POST /regression-check: old=5, new=2, threshold=2.
OUTPUT / OBSERVED RESULT	regression_detected=false; improvement_detected=true; result="No regression".
VALIDATION DETAIL	Improvement was explicitly recognized.
FINAL STATUS	PASS

Detailed Edge Cases - Continued

EC-09 - Huge slowdown - Situation: Current is many times baseline STATUS: PASS

EXPECTED BEHAVIOUR	Detect severe regression.
INPUT / TEST ACTION	POST /regression-check: old=1, new=100, threshold=2.
OUTPUT / OBSERVED RESULT	performance_ratio=100; regression_detected=true; result="Regression detected".
VALIDATION DETAIL	Severe slowdown detected.
FINAL STATUS	PASS

EC-10 - Temporary spike - Situation: One unusually slow execution STATUS: PASS

EXPECTED BEHAVIOUR	Avoid overreacting when using historical baseline logic.
INPUT / TEST ACTION	POST /regression-history-check: old=[1,1.1,0.9,1], new=[1,1,10], threshold=2.
OUTPUT / OBSERVED RESULT	Initial average-based logic returned regression=true (old_average=1, new_average=4). After robust median comparison update, retest returned regression_detected=false.
VALIDATION DETAIL	Single outlier no longer creates false regression.
FINAL STATUS	PASS

EC-11 - Noisy execution times - Situation: Times vary naturally STATUS: PASS

EXPECTED BEHAVIOUR	Avoid false positives.
INPUT / TEST ACTION	old=[1,1.1,0.9,1.05,0.95], new=[1.1,0.9,1.05,1,0.95], threshold=2.
OUTPUT / OBSERVED RESULT	regression_detected=false.
VALIDATION DETAIL	Natural noise did not trigger regression.
FINAL STATUS	PASS

EC-12 - Very small baseline - Situation: Baseline = 0.001 sec STATUS: PASS

EXPECTED BEHAVIOUR	Percentage-based comparison can become unstable; handle carefully.
INPUT / TEST ACTION	old=[0.001,0.0011,0.0009], new=[0.0012,0.0011,0.001], threshold=2.
OUTPUT / OBSERVED RESULT	regression_detected=false.
VALIDATION DETAIL	Small-baseline variation remained stable under the configured rule.
FINAL STATUS	PASS

Detailed Edge Cases - Continued

EC-13 - Missing current time - Situation: Current measurement unavailable STATUS: PASS

EXPECTED BEHAVIOUR	Do not make regression decision.
INPUT / TEST ACTION	POST /regression-history-check: old=[1,1.1,0.9], new=[], threshold=2.
OUTPUT / OBSERVED RESULT	Initial run: HTTP 500. After missing-current handling fix: current_measurement_available=false; regression_detected=null; result="No regression decision - current measurement unavailable".
VALIDATION DETAIL	Missing current measurement now returns an explicit no-decision result.
FINAL STATUS	PASS

EC-14 - Missing historical records - Situation: No baseline available STATUS: PASS

EXPECTED BEHAVIOUR	Record current result and establish baseline.
INPUT / TEST ACTION	POST /regression-history-check: old=[], new=[2], threshold=2.
OUTPUT / OBSERVED RESULT	baseline_created=true; baseline=2; regression_detected=false.
VALIDATION DETAIL	Current measurement established as baseline without regression claim.
FINAL STATUS	PASS

EC-15 - Very old historical data - Situation: Baseline is outdated STATUS: PASS

EXPECTED BEHAVIOUR	Handle baseline freshness according to project rules.
INPUT / TEST ACTION	old=[1,1.1,0.9], new=[3], threshold=2, baseline_age_days=365, baseline_max_age_days=30.
OUTPUT / OBSERVED RESULT	baseline_stale=true; baseline_created=true; baseline=3; regression_detected=false.
VALIDATION DETAIL	Stale baseline was replaced according to configured freshness limit.
FINAL STATUS	PASS

EC-16 - Query text changed - Situation: Current query differs from historical query STATUS: PASS

EXPECTED BEHAVIOUR	Treat as a different query/version or handle explicitly.
INPUT / TEST ACTION	historical_query_text="SELECT * FROM users;", current_query_text="SELECT id FROM users;", old=1, new=3, threshold=2.
OUTPUT / OBSERVED RESULT	query_changed=true; regression_detected=null; result="Different query/version - regression comparison not applied".
VALIDATION DETAIL	Changed query text bypassed invalid historical comparison.
FINAL STATUS	PASS

Detailed Edge Cases - Continued

EC-17 - Multiple executions at once - Situation: Concurrent query runs STATUS: PASS

EXPECTED BEHAVIOUR	Track each query/execution correctly.
INPUT / TEST ACTION	Three concurrent POST /regression-check requests via ThreadPoolExecutor(max_workers=3): Q1 1->1, Q2 1->3, Q3 2->2.
OUTPUT / OBSERVED RESULT	All three completed independently: Q1 false/No regression; Q2 true/Regression detected; Q3 false/No regression.
VALIDATION DETAIL	Concurrent requests returned separate correct results without cross-mixing.
FINAL STATUS	PASS

EC-18 - Database failure - Situation: Current execution cannot complete STATUS: PASS

EXPECTED BEHAVIOUR	Do not classify database failure itself as a performance regression.
INPUT / TEST ACTION	POST /regression-check with execution_failed=true, old=1, new=3, threshold=2.
OUTPUT / OBSERVED RESULT	execution_failed=true; regression_detected=null; result="Execution failed - not classified as performance regression".
VALIDATION DETAIL	Execution/database failure is separated from performance-regression classification.
FINAL STATUS	PASS

EC-19 - Configuration changes - Situation: Regression threshold changed STATUS: PASS

EXPECTED BEHAVIOUR	Apply currently configured rule consistently.
INPUT / TEST ACTION	Same 1->3 timing tested twice: threshold=2, then threshold=4.
OUTPUT / OBSERVED RESULT	threshold=2 -> regression_detected=true; threshold=4 -> regression_detected=false.
VALIDATION DETAIL	Changed configuration was applied consistently.
FINAL STATUS	PASS

EC-20 - Data size changed dramatically - Situation: Table grows significantly STATUS: PASS

EXPECTED BEHAVIOUR	Detect measured regression but distinguish cause from detection.
INPUT / TEST ACTION	POST /regression-check: data_size_changed=true, old=1, new=3, threshold=2.
OUTPUT / OBSERVED RESULT	regression_detected=true; regression_cause="Data size changed"; result="Regression detected".
VALIDATION DETAIL	Regression detection and cause attribution are separated.
FINAL STATUS	PASS

Edge Case Fix / Retest Notes

EC-01: Empty historical list initially caused HTTP 500. Added first-run baseline creation branch. Retest returned `baseline_created=true` and no regression.

EC-02: Zero historical baseline initially failed because the baseline return block was indented outside the intended condition. Function structure was corrected and a safe zero-baseline comparison returned no regression.

EC-10: Average-based history comparison overreacted to a single spike. Median-based robust comparison was added; retest returned `regression_detected=false`.

EC-13: Missing current measurement initially caused HTTP 500. Added explicit no-decision branch returning `regression_detected=null`.

EC-15: Added configurable `baseline_age_days` / `baseline_max_age_days` freshness handling. Stale baseline was replaced instead of used for a regression decision.

EC-16: Added historical/current query text version handling; changed text returns `query_changed=true` and no regression decision.

EC-18: Added `execution_failed` handling so execution/database failure is not classified as performance regression.

EC-20: Added `data_size_changed` and `regression_cause` so detection and cause attribution are distinct.

Final EC result: EC-01 through EC-20 = 20/20 PASS.

Evidence Appendix A - TC-01 to TC-20

The following pages reproduce the Swagger evidence captured during execution of the main regression test cases.

TC-01 - No performance change - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 1,
    "new_execution_time": 1,
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code	Details
200	Response body <pre>{ "old_execution_time": 1, "new_execution_time": 1, "regression_threshold": 2, "performance_ratio": 1, "regression_detected": false, "result": "No regression" }</pre> Response headers <pre>content-length: 155</pre>

Observed validation: HTTP 200; performance_ratio=1; regression_detected=false; result="No regression".

TC-02 - Small performance increase - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 1,
    "new_execution_time": 1.1,
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code	Details
200	Response body <pre>{ "old_execution_time": 1, "new_execution_time": 1.1, "regression_threshold": 2, "performance_ratio": 1.1, "regression_detected": false, "result": "No regression" }</pre> Response headers

Observed validation: HTTP 200; ratio=1.1; regression_detected=false; result="No regression".

TC-03 - Significant slowdown - PASS

Swagger request/response screenshot captured during the test execution.

```
'http://127.0.0.1:8000/regression-check' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "old_execution_time": 1,
  "new_execution_time": 5,
  "regression_threshold": 2
}'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code	Details
200	Response body <pre>{ "old_execution_time": 1, "new_execution_time": 5, "regression_threshold": 2, "performance_ratio": 5, "regression_detected": true, "result": "Regression detected" }</pre> Response headers <pre>content-length: 160 content-type: application/json date: Wed, 19 Aug 2026 12:07:00 GMT</pre>

Observed validation: HTTP 200; ratio=5; regression_detected=true; result="Regression detected".

TC-04 - Exactly at regression threshold - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 2,
    "new_execution_time": 4,
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code	Details
200	Response body <pre>{ "old_execution_time": 2, "new_execution_time": 4, "regression_threshold": 2, "performance_ratio": 2, "regression_detected": true }</pre>

Observed validation: HTTP 200; ratio=2; boundary rule applied as ratio >= threshold.

TC-05 - Slightly above threshold - PASS

Swagger request/response screenshot captured during the test execution.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 2,
    "new_execution_time": 4.1,
    "regression_threshold": 2
  }'
```

Request URL

`http://127.0.0.1:8000/regression-check`

Server response

Code

Details

200

Response body

```
{
  "old_execution_time": 2,
  "new_execution_time": 4.1,
  "regression_threshold": 2,
  "performance_ratio": 2.05,
  "regression_detected": true,
  "result": "Regression detected"
}
```

Response headers

Observed validation: HTTP 200; ratio=2.05; regression_detected=true.

TC-06 - Very large slowdown - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 0.5,
    "new_execution_time": 20,
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code

Details

200

Response body

```
{
  "old_execution_time": 0.5,
  "new_execution_time": 20,
  "regression_threshold": 2,
  "performance_ratio": 40,
  "regression_detected": true,
  "result": "Regression detected"
}
```

Response headers

```
content-length: 162
```

Observed validation: HTTP 200; ratio=40; regression_detected=true.

TC-07 - Query becomes faster - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 5,
    "new_execution_time": 2,
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code	Details
200	Response body <pre>{ "old_execution_time": 5, "new_execution_time": 2, "regression_threshold": 2, "performance_ratio": 0.4, "regression_detected": false, "result": "No regression" }</pre> Response headers

Observed validation: HTTP 200; ratio=0.4; regression_detected=false; result="No regression".

TC-08 - Multiple executions - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-history-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_times": [1, 2, 3],
    "new_execution_times": [2, 3, 4],
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-history-check
```

Server response

Code	Details
200	Response body <pre>{ "old_average": 2, "new_average": 3, "regression_threshold": 2, "performance_ratio": 1.5, "regression_detected": false, "result": "No regression" }</pre> Response headers <pre>content-length: 141</pre>

Observed validation: HTTP 200; old_average=2; new_average=3; ratio=1.5; regression_detected=false.

TC-09 - Same query repeated - PASS

Swagger request/response screenshot captured during the test execution.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-history-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_times": [1, 1, 1],
    "new_execution_times": [3, 3, 3],
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-history-check
```

Server response

Code	Details
200	Response body<pre>{ "old_average": 1, "new_average": 3, "regression_threshold": 2, "performance_ratio": 3, "regression_detected": true, "result": "Regression detected" }</pre> Response headers<pre>content-length: 146 content-type: application/json date: Wed, 19 Aug 2026 12:17:19 GMT server: uvicorn</pre>

Responses

Observed validation: HTTP 200; old_average=1; new_average=3; ratio=3; regression_detected=true.

TC-10 - Multiple queries - PASS

Swagger request/response screenshot captured during the test execution.

```
    },
    ],
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/multiple-regression-check
```

Server response

Code	Details
200	Response body <pre>{ "results": [{ "query_name": "Q1", "performance_ratio": 1, "regression_detected": false }, { "query_name": "Q2", "performance_ratio": 3, "regression_detected": true }, { "query_name": "Q3", "performance_ratio": 1, "regression_detected": false }] }</pre>

Observed validation: Q1 ratio=1 false; Q2 ratio=3 true; Q3 ratio=1 false.

TC-11 - Stable query - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-history-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_times": [2, 2.1, 2],
    "new_execution_times": [2.1, 2, 2.1],
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-history-check
```

Server response

Code	Details
200	Response body <pre>{ "old_average": 2.0333333333333333, "new_average": 2.0666666666666667, "regression_threshold": 2, "performance_ratio": 1.016393442622951, "regression_detected": false, "result": "No regression" }</pre>

Observed validation: old_average=2.0333; new_average=2.0667; ratio=1.01639; regression_detected=false.

TC-12 - Gradual slowdown - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-history-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_times": [1, 1.5, 2, 2.5],
    "new_execution_times": [5],
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-history-check
```

Server response

Code	Details
200	Response body <pre>{ "old_average": 1.75, "new_average": 5, "regression_threshold": 2, "performance_ratio": 2.857142857142857, "regression_detected": true, "result": "Regression detected" }</pre> Response headers

Observed validation: old_average=1.75; new_average=5; ratio=2.85714; regression_detected=true.

TC-13 - Sudden slowdown - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 1,
    "new_execution_time": 8,
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code	Details
200	Response body <pre>{ "old_execution_time": 1, "new_execution_time": 8, "regression_threshold": 2, "performance_ratio": 8, "regression_detected": true, "result": "Regression detected" }</pre>

Observed validation: HTTP 200; ratio=8; regression_detected=true.

TC-14 - Performance improvement after optimization - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_time": 8,
    "new_execution_time": 1,
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-check
```

Server response

Code	Details
200	Response body <pre>{ "old_execution_time": 8, "new_execution_time": 1, "regression_threshold": 2, "performance_ratio": 0.125, "regression_detected": false, "improvement_detected": true, "result": "No regression" }</pre> Response headers <pre>content-length: 185 content-type: application/json date: Wed, 19 Aug 2026 12:28:06 GMT server: uvicorn</pre>

Observed validation: Retest HTTP 200; ratio=0.125; regression_detected=false; improvement_detected=true; result="No regression".

TC-15 - Regression report - PASS

Swagger request/response screenshot captured during the test execution.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "query_name": "Q1",
    "old_execution_time": 1,
    "new_execution_time": 5,
    "regression_threshold": 2
  }'
```

Request URL

http://127.0.0.1:8000/regression-check

Server response

Code

Details

200

Response body

```
{
  "query_name": "Q1",
  "old_execution_time": 1,
  "new_execution_time": 5,
  "slowdown_percent": 400,
  "regression_threshold": 2,
  "performance_ratio": 5,
  "regression_detected": true,
  "improvement_detected": false,
  "result": "Regression detected"
}
```

Response headers

content-length: 232

Observed validation: HTTP 200; query_name=Q1; slowdown_percent=400; ratio=5; regression_detected=true.

TC-16 - Historical baseline available - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-history-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "old_execution_times": [1, 1.1, 0.9],
    "new_execution_times": [3],
    "regression_threshold": 2
  }'
```

Request URL

```
http://127.0.0.1:8000/regression-history-check
```

Server response

Code	Details
200	Response body <pre>{ "old_average": 1, "new_average": 3, "regression_threshold": 2, "performance_ratio": 3, "regression_detected": true, "result": "Regression detected" }</pre> Response headers

Observed validation: HTTP 200; old_average=1; new_average=3; ratio=3; regression_detected=true.

TC-17 - Different query plans - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "query_name": "Q1",
    "old_query_plan": "INDEX SCAN",
    "new_query_plan": "FULL TABLE SCAN",
    "old_execution_time": 1,
    "new_execution_time": 3,
    "regression_threshold": 2
  }'
```

Request URL

http://127.0.0.1:8000/regression-check

Server response

Code	Details
------	---------

200	
-----	--

Response body

```
{
  "query_name": "Q1",
  "old_query_plan": "INDEX SCAN",
  "new_query_plan": "FULL TABLE SCAN",
  "plan_changed": true,
  "old_execution_time": 1,
  "new_execution_time": 3,
  "regression_detected": true
}
```

3 29°C
Mostly cloudy



Search



Observed validation: HTTP 200; plan_changed=true; ratio=3; regression_detected=true.

TC-18 - Data growth causes slowdown - PASS

Swagger request/response screenshot captured during the test execution.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "query_name": "Q_DATA_GROWTH",
    "old_query_plan": "FULL TABLE SCAN",
    "new_query_plan": "FULL TABLE SCAN",
    "old_execution_time": 1,
    "new_execution_time": 3,
    "regression_threshold": 2
  }'
```

Request URL

http://127.0.0.1:8000/regression-check

Server response

Code

Details

200

Response body

```
{
  "query_name": "Q_DATA_GROWTH",
  "old_query_plan": "FULL TABLE SCAN",
  "new_query_plan": "FULL TABLE SCAN",
  "plan_changed": false,
  "old_execution_time": 1,
  "new_execution_time": 3,
  "slowdown_percent": 200,
  "regression_threshold": 2,
  "performance_ratio": 3,
  "regression_detected": true,
  "improvement_detected": false,
  "result": "Regression detected"
}
```

Observed validation: HTTP 200; plan_changed=false; slowdown_percent=200; ratio=3; regression_detected=true.

TC-19 - Index removed - PASS

Swagger request/response screenshot captured during the test execution.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "query_name": "Q_INDEX_REMOVED",
    "old_query_plan": "INDEX SCAN",
    "new_query_plan": "FULL TABLE SCAN",
    "old_execution_time": 1,
    "new_execution_time": 3,
    "regression_threshold": 2
  }'
```

Request URL

http://127.0.0.1:8000/regression-check

Server response

Code	Details
------	---------

200	
-----	--

Response body

```
{
  "query_name": "Q_INDEX_REMOVED",
  "old_query_plan": "INDEX SCAN",
  "new_query_plan": "FULL TABLE SCAN",
  "plan_changed": true,
  "old_execution_time": 1,
  "new_execution_time": 3,
  "slowdown_percent": 200,
  "regression_threshold": 2,
  "performance_ratio": 3,
  "regression_detected": true,
  "improvement_detected": false,
  "result": "Regression detected"
}
```

Response headers

content-length: 330



Search



Observed validation: HTTP 200; plan_changed=true; slowdown_percent=200; ratio=3; regression_detected=true.

TC-20 - Statistics / plan change - PASS

Swagger request/response screenshot captured during the test execution.

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/regression-check' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "query_name": "Q_STATS_PLAN_CHANGE",
    "old_query_plan": "INDEX SCAN",
    "new_query_plan": "FULL TABLE SCAN AFTER STATISTICS CHANGE",
    "old_execution_time": 1,
    "new_execution_time": 3,
    "regression_threshold": 2
  }'
```

Request URL

`http://127.0.0.1:8000/regression-check`

Server response

Code

Details

200

Response body

```
{
  "query_name": "Q_STATS_PLAN_CHANGE",
  "old_query_plan": "INDEX SCAN",
  "new_query_plan": "FULL TABLE SCAN AFTER STATISTICS CHANGE",
  "plan_changed": true,
  "old_execution_time": 1,
  "new_execution_time": 3,
  "slowdown_percent": 200,
  "regression_threshold": 2,
  "performance_ratio": 3,
  "regression_detected": true,
  "improvement_detected": false,
  "result": "Regression detected"
}
```

Response headers

Observed validation: HTTP 200; plan_changed=true; slowdown_percent=200; ratio=3; regression_detected=true.

Evidence Appendix B - EC-01 to EC-20

EC-01 includes its captured Swagger screenshot. EC-02 through EC-20 preserve the exact final response values / terminal evidence recorded in the testing conversation.

EC-01 - No historical data - PASS

Swagger evidence captured during the initial failing run; final retest values are recorded below.

Responses

Curl

```
curl -X 'POST' \
'http://127.0.0.1:8000/regression-history-check' \
-H 'accept: application/json' \
-H 'Content-Type: application/json' \
-d '{
  "old_execution_times": [],
  "new_execution_times": [1],
  "regression_threshold": 2
}'
```

Request URL

```
http://127.0.0.1:8000/regression-history-check
```

Server response

Code	Details
200	Response body <pre>{ "baseline_created": true, "baseline": 1, "regression_detected": false, "result": "No regression - baseline created" }</pre> Response headers <pre>content-length: 112 content-type: application/json date: Wed, 19 Aug 2026 13:54:11 GMT server: uvicorn</pre>

EC-01 - No historical data - PASS

Final retest evidence recorded after the targeted baseline fix.

Test input	POST /regression-history-check: old_execution_times=[], new_execution_times=[1], threshold=2.
Observed output	Initial run: HTTP 500. After baseline-empty handling fix: HTTP 200; baseline_created=true; baseline=1; regression_detected=false; result="No regression - baseline created".
Validation	First-time run now establishes baseline without claiming regression.
Final status	PASS

EC-02 - Historical baseline = 0 - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-history-check: old=[0], new=[1], threshold=2.
Observed output	After correcting return indentation/safe path: old_average=0.0; new_average=1.0; performance_ratio=1.0; regression_detected=false; result="No regression".
Validation	Zero baseline no longer crashes or divides by zero.
Final status	PASS

EC-03 - Negative execution time - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: old_execution_time=-1, new_execution_time=1, threshold=2.
Observed output	HTTP 422 validation response.
Validation	Negative execution time rejected by input validation.
Final status	PASS

EC-04 - Current time = baseline - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: old=2, new=2, threshold=2.
Observed output	HTTP 200; regression_detected=false; result="No regression".
Validation	No-change case correctly remained non-regression.
Final status	PASS

EC-05 - Current time slightly higher - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: old=1, new=1.1, threshold=2.
Observed output	HTTP 200; regression_detected=false; result="No regression".
Validation	Tiny variation did not create a false positive.
Final status	PASS

EC-06 - Exactly at threshold - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: old=2, new=4, threshold=2.
Observed output	HTTP 200; regression_detected=true; result="Regression detected".
Validation	Boundary consistently uses >= threshold.
Final status	PASS

EC-07 - Just above threshold - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: old=2, new=4.01, threshold=2.
Observed output	HTTP 200; regression_detected=true; result="Regression detected".
Validation	Just-above-threshold case detected correctly.
Final status	PASS

EC-08 - Query becomes faster - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: old=5, new=2, threshold=2.
Observed output	regression_detected=false; improvement_detected=true; result="No regression".
Validation	Improvement was explicitly recognized.
Final status	PASS

EC-09 - Huge slowdown - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: old=1, new=100, threshold=2.
Observed output	performance_ratio=100; regression_detected=true; result="Regression detected".
Validation	Severe slowdown detected.
Final status	PASS

EC-10 - Temporary spike - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-history-check: old=[1,1.1,0.9,1], new=[1,1,10], threshold=2.
Observed output	Initial average-based logic returned regression=true (old_average=1, new_average=4). After robust median comparison update, retest returned regression_detected=false.
Validation	Single outlier no longer creates false regression.
Final status	PASS

EC-11 - Noisy execution times - PASS

Recorded execution evidence from the testing conversation.

Test input	old=[1,1.1,0.9,1.05,0.95], new=[1.1,0.9,1.05,1,0.95], threshold=2.
Observed output	regression_detected=false.
Validation	Natural noise did not trigger regression.
Final status	PASS

EC-12 - Very small baseline - PASS

Recorded execution evidence from the testing conversation.

Test input	old=[0.001,0.0011,0.0009], new=[0.0012,0.0011,0.001], threshold=2.
Observed output	regression_detected=false.
Validation	Small-baseline variation remained stable under the configured rule.
Final status	PASS

EC-13 - Missing current time - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-history-check: old=[1,1.1,0.9], new=[], threshold=2.
Observed output	Initial run: HTTP 500. After missing-current handling fix: current_measurement_available=false; regression_detected=null; result="No regression decision - current measurement unavailable".
Validation	Missing current measurement now returns an explicit no-decision result.
Final status	PASS

EC-14 - Missing historical records - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-history-check: old=[], new=[2], threshold=2.
Observed output	baseline_created=true; baseline=2; regression_detected=false.
Validation	Current measurement established as baseline without regression claim.
Final status	PASS

EC-15 - Very old historical data - PASS

Recorded execution evidence from the testing conversation.

Test input	old=[1,1.1,0.9], new=[3], threshold=2, baseline_age_days=365, baseline_max_age_days=30.
Observed output	baseline_stale=true; baseline_created=true; baseline=3; regression_detected=false.
Validation	Stale baseline was replaced according to configured freshness limit.
Final status	PASS

EC-16 - Query text changed - PASS

Recorded execution evidence from the testing conversation.

Test input	historical_query_text="SELECT * FROM users;"; current_query_text="SELECT id FROM users;"; old=1, new=3, threshold=2.
Observed output	query_changed=true; regression_detected=null; result="Different query/version - regression comparison not applied".
Validation	Changed query text bypassed invalid historical comparison.
Final status	PASS

EC-17 - Multiple executions at once - PASS

Recorded execution evidence from the testing conversation.

Test input	Three concurrent POST /regression-check requests via ThreadPoolExecutor(max_workers=3): Q1 1->1, Q2 1->3, Q3 2->2.
Observed output	All three completed independently: Q1 false/No regression; Q2 true/Regression detected; Q3 false/No regression.
Validation	Concurrent requests returned separate correct results without cross-mixing.
Final status	PASS

EC-18 - Database failure - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check with execution_failed=true, old=1, new=3, threshold=2.
Observed output	execution_failed=true; regression_detected=null; result="Execution failed - not classified as performance regression".
Validation	Execution/database failure is separated from performance-regression classification.
Final status	PASS

EC-19 - Configuration changes - PASS

Recorded execution evidence from the testing conversation.

Test input	Same 1->3 timing tested twice: threshold=2, then threshold=4.
Observed output	threshold=2 -> regression_detected=true; threshold=4 -> regression_detected=false.
Validation	Changed configuration was applied consistently.
Final status	PASS

EC-20 - Data size changed dramatically - PASS

Recorded execution evidence from the testing conversation.

Test input	POST /regression-check: data_size_changed=true, old=1, new=3, threshold=2.
Observed output	regression_detected=true; regression_cause="Data size changed"; result="Regression detected".
Validation	Regression detection and cause attribution are separated.
Final status	PASS

Final Conclusion

Validation set	Passed	Failed	Result
TC-01 to TC-20	20	0	PASS
EC-01 to EC-20	20	0	PASS
TOTAL	40	0	PASS

Final result: 40/40 PASS, 0 FAIL. All requested functional test cases and edge cases have a final PASS status in the current testing record after the documented fixes and retests.

Evidence integrity: This report distinguishes screenshot evidence from text/terminal evidence and does not invent missing screenshots.